Data Analytics 1

Assignment 4

Release: 29-10-2026 Deadline: 15-11-2026

Section 1: Classification

An automobile company plans to enter new markets with its existing products. In its existing market, the sales team has classified customers into four segments (A, B, C, D) and follows a different outreach strategy for each segment. The company believes the new market behaves similarly and wants to predict the segment of each of its 789 potential new customers.

In this section, you are required to build multiclass classifiers that classify customers based on a mix of numerical, nominal and ordinal attributes. Use the data in `Customer_train.csv` to train your classifiers for the column `Segmentation` (target variable). `Customer_test.csv` contains the potential customers in the same format, without the target.

Note: You may use sklearn or other libraries for the metrics, for binary SVMs (Task 1) and for decision trees (Task 2). You may not use `OneVsOneClassifier` or `OneVsRestClassifier` from `sklearn.multiclass`, or the built-in multiclass mode of `SVC`/`LinearSVC` (Task 1). Every SVM must be trained on a binary target. You may not use `RandomForestClassifier`, `ExtraTreesClassifier` or `BaggingClassifier` from `sklearn.ensemble` (Task 2).

Task 1: One-vs-One and One-vs-All SVM Classifiers

Build two multiclass classifiers from binary SVMs, one-vs-one and one-vs-all, and train them on `Customer_train.csv`. Handle missing values, numerical vs. nominal vs. ordinal attributes (e.g. `Spending_Score`: Low < Average < High), and class imbalance. Learn all preprocessing from the training data only.

a. **One-vs-one:** train $\binom{K}{2}$ binary SVMs, one for each pair of classes, using only the rows of those two classes. Predict by majority voting, and use an explicit tie-breaking rule that does not depend on the order of the class labels. Running `teamId_classifier_ovo.py <path to test file>` (the test file will be in the same format) should output an `ovo.csv` file with the predicted labels in a column named `predicted` (all lower case), one row per test row, in the same order as the input.

b. **One-vs-all:** train K binary SVMs, each separating one class from all the others, and predict the class with the highest decision score. Running `teamId_classifier_ova.py <path to test file>` should output an `ova.csv` file in the same format as above.

c. Select the kernel, C, γ and class weights using stratified k-fold cross-validation ($k \geq 5$) on the training data. Report the search grid and the cross-validated mean and standard deviation of each configuration.

Task 2: Random Forest Classifier

Build a random forest classifier using n decision trees for the above dataset.

a. Implement bootstrap sampling of rows, random feature sub-sampling at every split, and aggregation of tree predictions yourself. Also compute the out-of-bag (OOB) accuracy.

b. Try $n \in \{1, 5, 10, 25, 50, 100, 200\}$ for at least two values of `max_depth`, one of which is unrestricted. Plot OOB accuracy and 5-fold cross-validation accuracy against n, and explain the shape of the curves.

c. Print the predictions for the test dataset in the notebook and save them as `rf.csv` (column `predicted`). Print the accuracy, the per-class precision, recall and F1-score, and the macro-F1 using cross-validation or OOB predictions.

d. Compute permutation feature importance on held-out data. Which attributes drive the segmentation, and do they make business sense?

Task 3: Report and Analysis

Evaluate all three models (one-vs-one, one-vs-all, random forest) on the same cross-validation folds.

a. How does class imbalance affect multiclass classification, and how does it interact with the one-vs-all decomposition? What strategies can be employed to mitigate its impact, especially with small datasets? Support at least one claim with an experiment.

b. How can the choice of hyperparameters (C, γ and the kernel for SVM; number of trees, `max_depth`, `min_samples_leaf` and `max_features` for random forest) make the classifiers more prone to over- or under-fitting? Show a validation curve (training vs. cross-validation score) for γ and for `max_depth`.

c. Plot the confusion matrix of each model and report precision, recall and F1-score per class, along with macro- and weighted-F1. Which segments are most often confused with each other?

d. Compare one-vs-one and one-vs-all. Report the difference in accuracy with a 95% bootstrap confidence interval and perform McNemar's test on the paired predictions. Is the difference statistically significant? Which approach would you choose for this dataset, and why?

e. Report the accuracy and macro-F1 of a majority-class baseline and a stratified-random baseline. Is your best model significantly better than the majority baseline (use a one-sided binomial test or a label-permutation test)? Is the improvement practically meaningful? In at most 150 words, advise the company's sales team on how far they should rely on the predicted segments for the 789 potential customers.

Include all the above plots, analysis and answers to the theoretical questions in `teamId_report_1.pdf`.

Section 2: Clustering

Introduction

This section is on clustering and data visualization. The goal is to gain experience with common clustering techniques and data preprocessing/visualization tools. There are three parts in this section:

- The first part involves data visualization and preprocessing using PowerBI.
- The second part requires implementing K-Means from scratch.
- The third part requires comparing different clustering algorithms. In this part, sklearn can be used for implementations, including for K-Means.

Task 1: Data Visualization and Preprocessing Using PowerBI

Use `Dataset1/` for this part. This dataset contains data for approximately 2,240 customers of a retailer. Each row holds details about one customer: personal information, purchasing behavior, and interactions with marketing campaigns.

- **Data exploration and visualization:** import the dataset into PowerBI and build an interactive report with slicers (e.g. Education, Marital Status, Income band). It should include:
  - bar charts showing the distribution of categorical variables (e.g. Education, Marital Status);
  - histograms showing the distribution of numerical variables (e.g. Age, Income);
  - scatter plots of relationships between variables (e.g. Income vs. Total Spending);
  - a correlation matrix or heatmap;
  - at least two DAX measures (e.g. Total Spending, Campaign Response Rate).
- **Data preprocessing:** using Power Query, perform and document the following, reporting the number of rows or values each step affects:
  - handle missing values (e.g. `Income`);
  - handle implausible values and outliers (e.g. very old birth years, extreme incomes);
  - make inconsistent categories consistent (e.g. `Marital_Status` values such as `Alone`, `Absurd`, `YOLO`);
  - create derived features such as Age, Customer Tenure, Total Spending, Total Purchases and Total Children. Compute Age and Tenure relative to a fixed reference date within the data rather than today's date, and explain why.

  Export the cleaned dataset as `teamId_marketing_cleaned.csv`.
- **Insights:** provide at least five specific, quantified insights based on the visualizations: trends, patterns or anomalies in the data, and what each implies for the business.

Task 2: K-Means Clustering Implementation

Use the preprocessed dataset from Task 1 for this part. Do not use `Response`, `AcceptedCmp1`–`AcceptedCmp5`, or any feature derived from them as clustering features. They are used in Section 3.

- Implement the K-Means clustering algorithm without using any built-in clustering functions. Include k-means++ initialization, a convergence criterion, handling of empty clusters, and multiple random restarts. Verify your implementation against `sklearn.cluster.KMeans` by comparing inertia and the adjusted Rand index (ARI) between the two labelings.

- Justify your choice of features and scaling. Why is K-Means on unscaled or one-hot encoded features problematic?
- Run your implementation for $k = 2, 3, \ldots, 10$. Use the elbow method, the silhouette score and cluster stability (ARI across random seeds or bootstrap samples) to choose the number of clusters. Discuss how usable the chosen number of segments is from a business point of view.
- Analyze the clusters formed and interpret them in the context of customer segmentation. Profile each cluster in original units and give it a descriptive persona. Test whether `Education` and `Complain`, which were not used for clustering, differ across clusters (χ^2 test and Cramér's V). Why do significance tests on the clustering features themselves prove nothing?

Task 3: Clustering with Different Algorithms

Use the datasets in the `Dataset2/` folder for this part.

- For the four datasets in this part, use the following clustering algorithms:
  - K-Means
  - Agglomerative Clustering (compare single, complete, average and Ward linkage, and plot a dendrogram for at least one dataset)
  - DBSCAN
- Perform the following tasks:
  1. Visualize each dataset, then apply each clustering algorithm to each dataset and visualize the clusters using a scatter plot (a different color for each cluster, noise points in grey).
  2. Choose the number of clusters using the elbow method or the silhouette score. For DBSCAN, choose `eps` using a k-distance plot (with $k =$ `min_samples`) for each dataset and justify your choice of `min_samples`. Report the parameters used.
  3. Analyze which algorithm performs best for each dataset. Where an algorithm fails, explain the reason in terms of its assumptions: cluster shape, cluster size and variance, chaining, and density.
  4. Tabulate the silhouette score, Davies–Bouldin index and Calinski–Harabasz index for each dataset and each algorithm. Explain why the sum of intra-cluster distances cannot fairly compare algorithms that produce different numbers of clusters, and why the silhouette score can favor a convex but incorrect clustering.
  5. Ignore the noise points predicted by DBSCAN when computing its scores, and report the number and percentage of noise points in your table. Discuss how ignoring noise points can inflate DBSCAN's scores.
- Plot the visualizations in your notebook and write the reasoning and inferences. Tabulate the scores in notebook markdown.

Task 4: Report and Analysis

Write a comprehensive report, `teamId_report_2.pdf`, detailing the results and conclusions from all parts of this section. Include:

- the plots and insights from all parts;
- a reflection on using PowerBI vs. Python for data visualization and preprocessing.

Use insights from both the visualizations and the clustering to reach your conclusions.

Section 3: Case Study

Turning a Loss-Making Campaign Around

The retailer in `Dataset1/` ran a pilot campaign offering a new product to its entire customer base. The column `Response` records whether each customer accepted the offer. Each contact cost the company 3 monetary units (MU), and each acceptance brought in 11 MU of revenue. These are the constant columns `Z_CostContact` and `Z_Revenue` in the raw data. The Chief Marketing Officer (CMO) must now decide whether to run a second campaign and, if so, which customers to contact.

Use `teamId_marketing_cleaned.csv` and the clusters from Section 2, Task 2.

a. **Diagnosing the pilot.**

i. Compute the response rate of the pilot campaign with a 95% Wilson confidence interval, and its total cost, revenue and profit.

ii. Derive the break-even response probability p^* above which contacting a customer has a positive expected profit. Explain why contacting every customer lost money.

b. **Who responds?**

i. Test whether `Response` is associated with at least six customer attributes, at least two of them categorical (e.g. Education, Marital Status, children at home, Income, Recency, Total Spending, web visits). Use χ^2 tests for categorical attributes, and Welch's t-test or the Mann–Whitney U test for numerical attributes; justify your choice.

ii. Report an effect size for every test (Cramér's V, Cohen's d or rank-biserial correlation), and adjust for multiple comparisons using the Holm–Bonferroni method.

iii. Which findings are statistically significant, and which are practically important? Which two attributes would you present to the CMO, and why?

c. **Segment-level view.**

i. For each cluster from Section 2, report its size, its response rate with a 95% confidence interval, and the expected profit if only that segment were contacted.

ii. Test whether the response rate differs across clusters.

iii. Would contacting only the most profitable segments beat contacting everyone? By how much?

d. **Response model and targeting.**

i. For every feature, including the ones you engineered in PowerBI, check whether it would be available *before* the next campaign is sent. Identify any feature that leaks the target, show its effect on the cross-validated ROC-AUC, and remove it.

ii. Build a logistic regression model and one of the models from Section 1 (SVM or random forest; library implementations are allowed here) to predict `Response`. Use stratified 5-fold cross-validation with all preprocessing inside the folds. Report ROC-AUC and PR-AUC, and compare the PR-AUC with that of a random classifier.

iii. Using out-of-fold predicted probabilities, plot a cumulative gains (lift) chart and a profit curve (expected profit vs. fraction of customers contacted). Find the profit-maximizing threshold and compare it with p^* from part (a). Explain any difference in terms of probability calibration.

iv. Compare the expected profit, number of contacts and number of responders under four policies: (1) contact everyone, (2) contact no one, (3) contact only the best segments from part (c), and (4) your model-based policy. Give a 95% bootstrap confidence interval for the profit of your recommended policy.

e. **Recommendation to the CMO.** Write a memo of at most 2 pages for a non-technical reader. It should cover:

   i. your recommended decision, with the expected profit and its uncertainty;

   ii. which customers to target, described as customer personas;

   iii. the key risks and assumptions behind your recommendation;

   iv. why a model of *who responds* differs from a model of *who responds because of the campaign*, and how a randomized control (hold-out) group in the next campaign would measure its incremental profit;

   v. one ethical or privacy concern with targeting customers on attributes such as marital status or age, and how you would address it.

Include the analysis in `teamId_case_study.ipynb` and the memo in `teamId_case_study_memo.pdf`.

Submission Instructions

Submit a `teamId_assignment4.zip` file containing the following:

- `teamId_classifier_ovo.py`
- `teamId_classifier_ova.py`
- `teamId_random_forest.ipynb`
- `teamId_report_1.pdf`
- `teamId_clustering_Part1.pbix` (PowerBI file for Section 2, Task 1)
- `teamId_marketing_cleaned.csv`
- `teamId_clustering_Part2.ipynb` (Jupyter Notebook for Section 2, Task 2)
- `teamId_clustering_Part3.ipynb` (Jupyter Notebook for Section 2, Task 3)
- `teamId_report_2.pdf`
- `teamId_case_study.ipynb`
- `teamId_case_study_memo.pdf`
- Any intermediate files used